# Abdelmalek Essadi University

*Algebra IV — Make-up Exam*

*Year: 2021*

## Exercise 1:

Let $a \in \mathbb{R}$. Consider the matrix

$$A_a = \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R}).$$

1) Determine the characteristic polynomial of the matrix $A_a + I_3$.
2) For which real values of $a$ is the matrix $A_a$ diagonalizable?
3) Write the minimal polynomial of $A_a$.
4) Choose $a = 1$ and set $A = A_1$.
   1) Determine a diagonalization relation of $A$.
   2) Let $n \in \mathbb{N}^*$. Consider the polynomial $P(X) \in \mathbb{R}[X]$ defined by $P(X) = (X + I)^n$. Determine the matrix $P(A)$.

**Answer Area**

(1) Characteristic polynomial of $A_a + I_3$:

$$\chi_{A_a+I_3}(X) = X\left(X^2 - 3X - a + 1\right).$$

(2) $A_a$ is diagonalizable over $\mathbb{R}$ iff $a > -\frac{5}{4}$. If $a = -\frac{5}{4}$ it is not diagonalizable (double eigenvalue). If $a < -\frac{5}{4}$ eigenvalues are complex.

(3) Minimal polynomial:

- If $a > -\frac{5}{4}$: $(X + 1)(X - \lambda_1)(X - \lambda_2)$.
- If $a = -\frac{5}{4}$: $(X + 1)\left(X - \frac{1}{2}\right)^2$.
- If $a < -\frac{5}{4}$: $(X + 1)\left(X^2 - X - (1 + a)\right)$.

(4) For $a = 1$, $A = A_1$:

- Eigenvalues: $-1$ (double), $2$.
- Diagonalization: $A = P\,\mathrm{diag}(-1, -1, 2)\,P^{-1}$ with eigenvectors

$$v_1 = (-1, 1, 0)^T,\ v_2 = (-1, 0, 1)^T,\ v_3 = (1, 1, 1)^T.$$

$$- \ (A+I)^n = 3^{n-1} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

# Exercise 2:

Consider the matrix $A \in M_3(\mathbb{R})$ defined by

$$A = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 3 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Determine a Jordan reduction of the matrix $A$.

**Answer Area**

- Characteristic polynomial: $(X-1)^3$.
- $\dim\ker(A-I) = 2$, so the Jordan form has one block of size 2 and one block of size 1.
- Jordan form:
$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$
- One Jordan chain: eigenvector $u_1 = (1,1,0)^T$ and generalized vector $w = (-\frac{1}{2}, 0, 0)^T$ s.t. $(A-I)w = u_1$.

# Exercise 3:

Let $A \in M_n(\mathbb{C})$ be invertible and with $n$ distinct eigenvalues. A square root of the matrix $A$ is any matrix $M \in M_n(\mathbb{C})$ satisfying $M^2 = A$.

Show that the matrix $A$ has $2^n$ square roots, all of which are polynomials in $A$.

**Answer Area**

- Since $A$ has $n$ distinct eigenvalues, it is diagonalizable: $A = P\,\mathrm{diag}(\lambda_i)\,P^{-1}$.
- A square root must be of the form $M = P\,\mathrm{diag}(\mu_i)\,P^{-1}$ with $\mu_i^2 = \lambda_i$.
- Each $\lambda_i$ has two square roots, giving exactly $2^n$ matrices $M$.
- By Lagrange interpolation, there exists a polynomial $p$ such that $p(\lambda_i) = \mu_i$.
- Therefore $M = p(A)$, so every square root is a polynomial in $A$.